

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization.(BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 5

Total Marks: 50

Annexure A

Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Code of Conduct:

I ALLAPAREDDY JAHNAVI (192472214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

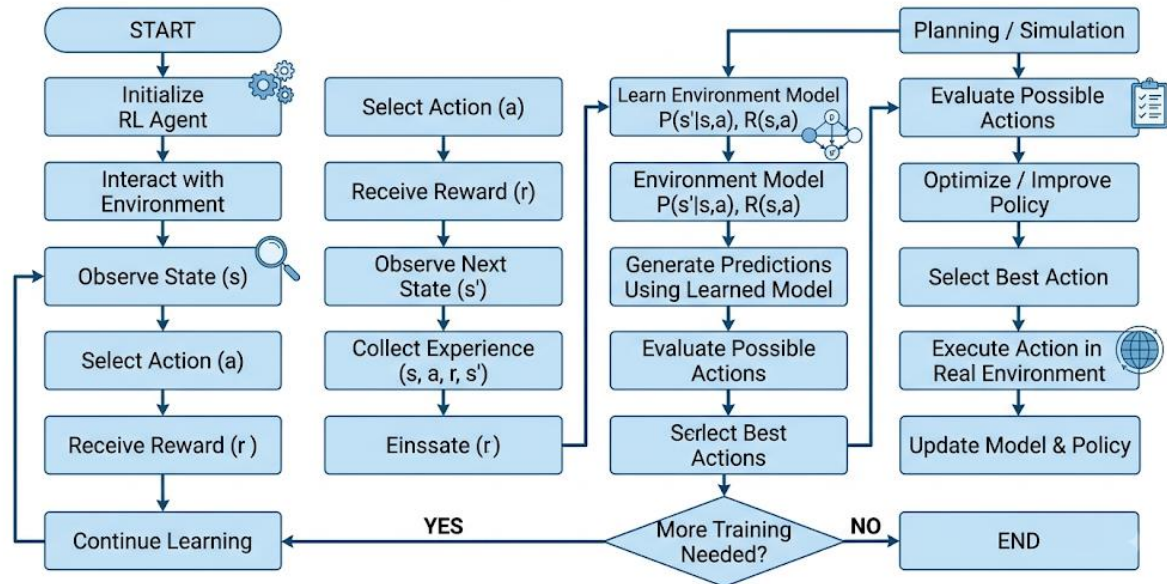
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Jahnavi

Question 1 – Model-Based RL Approach

Q. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

Model-Based Reinforcement Learning (MBRL) Process Flow



Explanation:

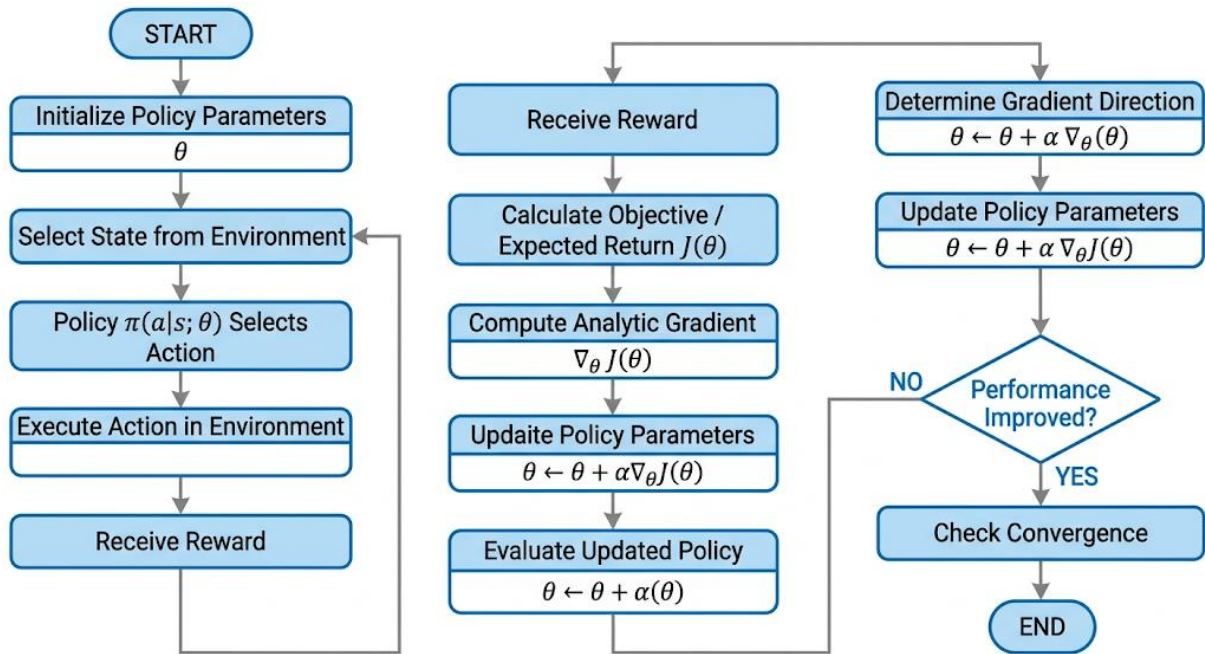
Model-Based Reinforcement Learning first collects real experience from the environment and uses it to learn an approximate model of the environment. The learned model predicts the next state and reward for different actions. A planning module then uses these predictions to evaluate possible future actions. The policy is optimized based on the predicted outcomes, and the improved policy is executed in the real environment. This process continues iteratively until the policy reaches satisfactory performance.

Question 2 – Analytic Gradient Computation

Q. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

Explanation:

Analytic gradient computation calculates how a change in policy parameters affects the expected reward. The gradient ($\nabla_{\theta} J(\theta)$) indicates the direction in which the policy parameters should be changed to improve performance.



An optimization rule such as gradient ascent is then used:

$$\theta_{\text{new}} = \theta + \alpha \nabla_{\theta} J(\theta)$$

where θ represents policy parameters and α represents the learning rate. This process is repeated until the policy converges to a satisfactory solution.

Question 3 – Sampling-Based Planning

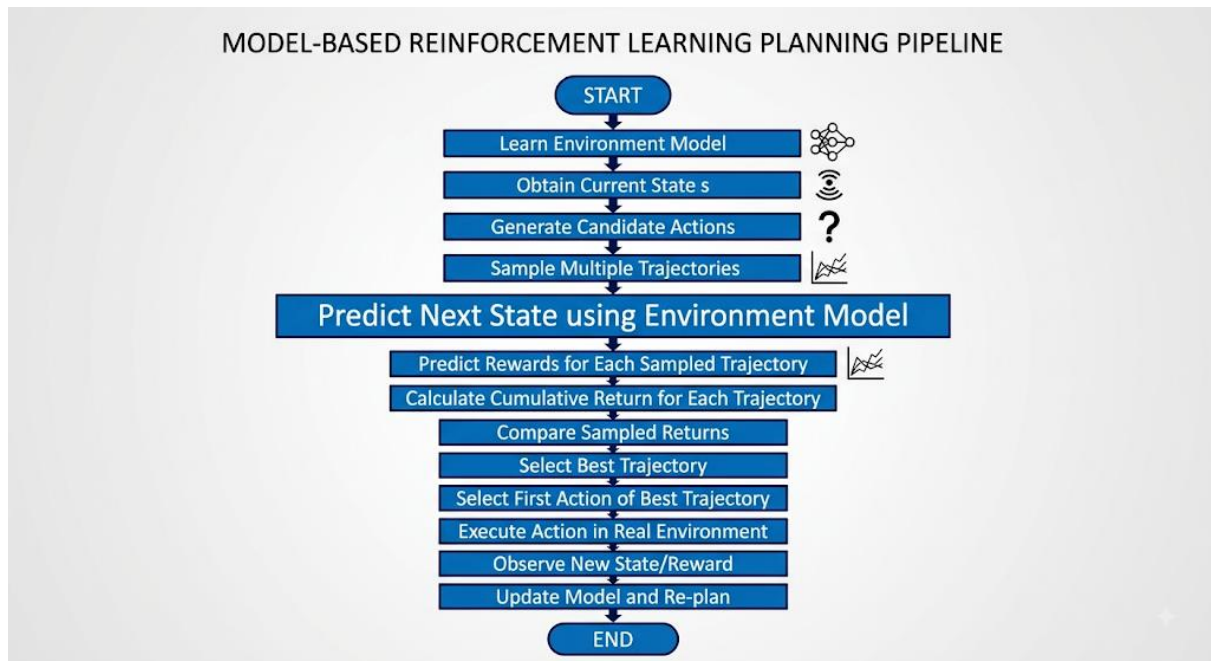
Q. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

Explanation:

Sampling-based planning avoids evaluating every possible future trajectory. Instead, the planner samples a number of possible action sequences using the learned environment model. Each trajectory is simulated, and its expected cumulative reward is calculated. The trajectory with the highest predicted return is selected, and its first action is executed in the real environment. The process is then repeated from the new state.

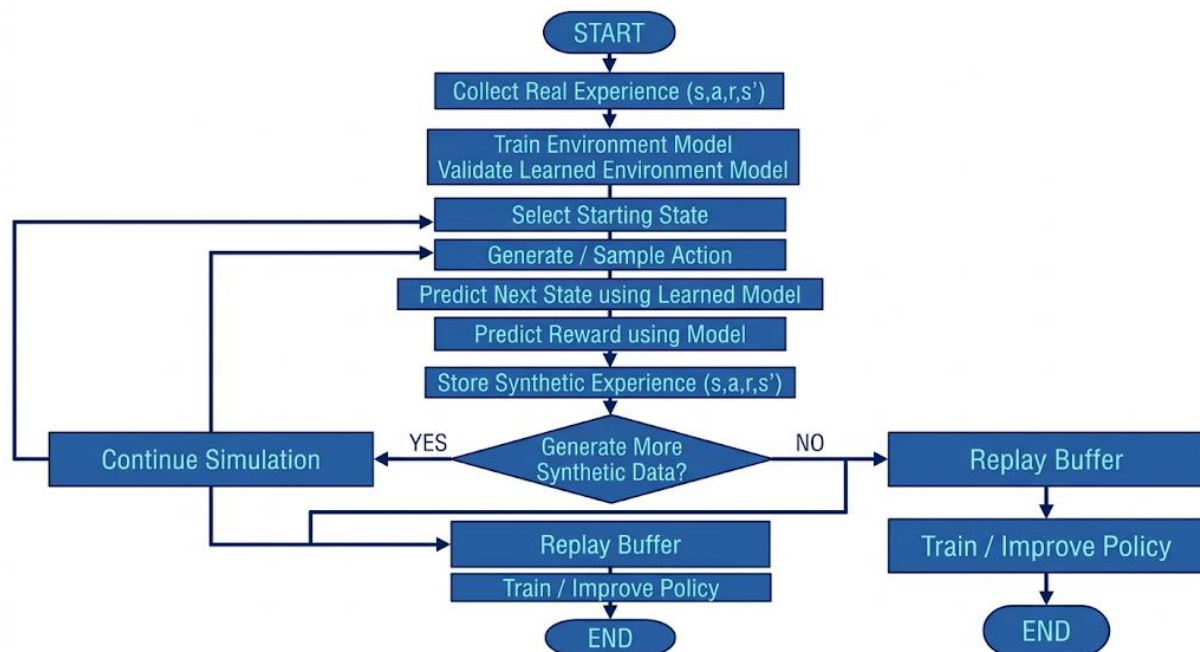
A simplified return calculation is: $G_t = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots = \sum_{k=0}^{\infty} \gamma^k r_{k+1}$

where G is the cumulative return and γ is the discount factor.



Question 4 – Model-Based Data Generation

Q. Draw a flowchart illustrating synthetic data generation using a learned environment model.



Explanation:

Model-Based RL can generate additional training experiences without repeatedly interacting with the real environment. After learning the environment dynamics from real data, the model

is used as a simulator. Starting from a state, actions are sampled and the model predicts the next state and reward. These synthetic experiences are stored in a replay buffer and used to improve the policy.

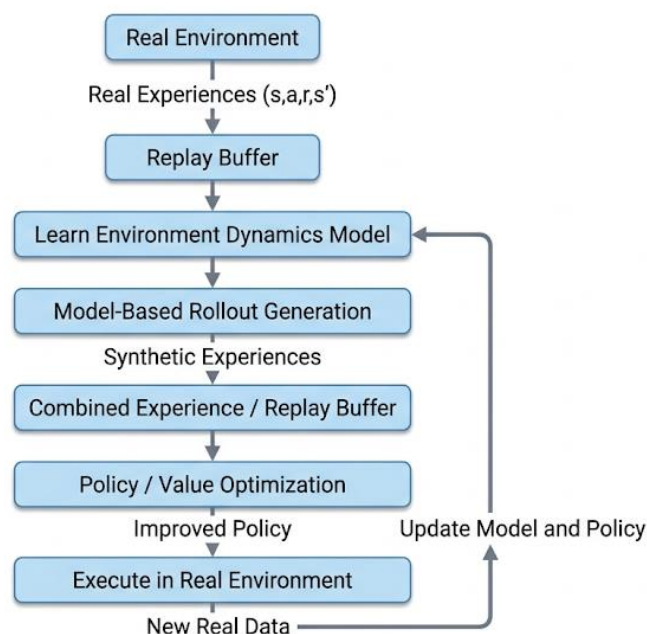
Advantages:

- Reduces interaction with the real environment.
- Improves sample efficiency.
- Produces additional training experiences.
- Can support faster policy learning.

Limitation: If the learned model is inaccurate, synthetic experiences may contain errors and negatively affect policy learning.

Question 5 – Value-Equivalence Prediction and Policy Optimization

Q. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.



Explanation of MBPO

Model-Based Policy Optimization (MBPO) combines a learned environment model with policy optimization. The agent first collects real experiences from the environment and stores them in a replay buffer. These experiences are used to learn an approximate dynamics model.

The learned model generates short synthetic rollouts from real states. These model-generated experiences are combined with real experiences and used to train the policy and value functions. The improved policy is then executed in the real environment, generating new data that further improves the model.

Main Components

Component	Function
Real Environment	Provides actual states, actions and rewards
Replay Buffer	Stores real and generated experiences
Dynamics Model	Predicts next states and rewards
Model Rollouts	Generates synthetic training data
Policy	Selects actions
Value Function	Estimates expected future rewards
Policy Optimization	Improves the policy using collected experiences

Overall MBPO Process:

Real Data → Learn Model → Generate Short Rollouts → Policy Optimization → Improved Policy → Real Environment

Result: MBPO improves sample efficiency by using a learned model to generate additional experiences while limiting the effect of model errors through short model-based rollouts.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness